INT396 · UNSUPERVISED LEARNING

UNIT 2

Partition-Based Clustering

How a computer sorts things into groups when nobody has told it what the groups are.

What clustering is • Hard vs Soft • k-Means • Initialisation • k-Medoids • Scaling • MiniBatch • Checking the answer

UNIT II ROADMAP

What This Unit Covers

UNIT 2

Start from the problem. Add one algorithm. Then find out where it fails and what to do about it.

- First - what clustering even is
  - The idea, and what it assumes about your data
- Then - the main algorithm, k-Means
  - How it works, how it starts, and where it breaks
- Then - two variants that fix its weaknesses
  - k-Medoids for messy data, MiniBatch for big data
- Last - how you check the answer
  - Because there is no marked answer sheet to compare against

Fundamentals
what clustering is

k-Means
the main algorithm

Variants
k-Medoids, MiniBatch

Validation
checking the answer

01

SECTION

What Clustering Is

IN THIS SECTION

- The problem, before any algorithm
- What the computer is given, and what it is not
- Hard clustering and soft clustering

2.1 WHAT CLUSTERING IS

Start With the Problem

KEY CONCEPT · **A shop owner has a list of 5,000 customers**

What they have how often each customer visits, and how much each one spends.

What they want to treat different kinds of customer differently.

What they do NOT have any column saying what kind of customer anyone is. Nobody has ever labelled them.

You Already Do This

REAL-WORLD APPLICATION · **Sorting a drawer of loose socks**

Nobody hands you categories you are not told 'these are the sports socks'. You just look at what is there.

You group by what you can see colour, length, thickness - whatever makes some socks look like each other.

The groups come from the socks not from a rulebook. That is exactly what clustering does.

So: What Is Clustering?

UNIT 2

Numbers in, groups out - and nothing in between tells it what the groups should be.

INPUT

A table of numbers

- Rows are things - customers, socks, cells
- Columns are measurements about them
- No column says what group anything belongs to

TASK

Put similar rows together

- Rows in the same group should resemble each other
- Rows in different groups should not

OUTPUT

A group label per row

- Every row gets exactly one group
- No row is left out, no group is empty
- That is what 'partition' means

The Hard Part: There Is No Right Answer

UNIT 2

COMMON PITFALL · **This is the biggest difference from anything you have done before**

In a normal exercise you get a question and a correct answer, and you can check yourself against it.

Here nobody knows the correct grouping. There is nothing to check against.

So you cannot ask 'is this clustering CORRECT?' You can only ask 'is it tidy?' and 'is it useful?'

THINK If there is no right answer, how would you argue a clustering is a bad one?

Four Things It Quietly Assumes

UNIT 2

Break any one of these and you still get an answer. It is just a wrong one.

ASSUMPTION 1

Groups actually exist

- Ask for 3 groups and you get 3 groups
- Even from data with no structure at all

ASSUMPTION 2

You know how many

- The number of groups is your input
- The algorithm never works it out for you

ASSUMPTION 3

Groups are round blobs

- Because it groups by 'nearest centre'
- A crescent-shaped group cannot be found

ASSUMPTION 4

Features are comparable

- It adds up differences across columns
- So a big-numbered column takes over

PREDICT Which of the four do you think is broken most often in real work?

Hard Clustering: Pick One

REAL-WORLD APPLICATION · **Like being sorted into a house at school**

The rule every student is in exactly one house. No exceptions.

What you get one label per row. Customer 41 is in group B.

Why it is useful you can act on it. One customer, one marketing campaign.

Soft Clustering: Give a Percentage

REAL-WORLD APPLICATION · **Like a student who is 60% science, 40% arts**

The rule every row gets a share in every group, and the shares add up to 1.

What you get customer 41 is 0.7 group B, 0.2 group A, 0.1 group C.

Why it is useful you can see who sits on a boundary - and boundary cases are where decisions go wrong.

Hard vs Soft - Which Do You Want?

The only question is whether a row is allowed to belong to more than one group.

	Hard clustering	Soft clustering
Each row gets	One group label	A share in every group
Boundary cases	Invisible - forced to pick a side	Visible - you can see the split
Easy to act on?	Yes - one row, one action	Less so - what do you do with 60/40?
Used in this unit	k-Means, k-Medoids, MiniBatch	Mentioned for contrast only

02

SECTION

k-Means:
The Main Algorithm

IN THIS SECTION

- What makes one grouping better than another
- The two steps, repeated
- How it starts, and why that matters
- Where it breaks

First Question: What Makes a Grouping GOOD?

UNIT 2

KEY CONCEPT · **Before we can find a good grouping we must define one**

The instinct members of a group should be close to each other.

Turn it into something measurable give each group a centre, then ask how far its members sit from it.

Small total means tight groups. Large total means loose ones.

Picture a Wedding Hall

REAL-WORLD APPLICATION · **Guests seated at round tables**

Good seating everybody is sitting close to their own table.

Bad seating people are scattered, sitting far from the table they were assigned to.

Measure it add up how far every guest is from their own table's centre. Small total, good seating.

That Total Has a Name: Inertia

One number that says how tight a whole grouping is.

WHAT IT IS

One number for the whole grouping

- Total distance from every point to its own centre
- Also called WCSS, or 'within-cluster sum of squares'

WHAT IT DOES

Lets you compare

- Grouping A scores 240. Grouping B scores 190.
- B is tighter. Now it is a fact, not an opinion.

THE WHOLE JOB

Make it small

- k-Means hunts for the grouping with the lowest inertia
- Everything it does follows from that

One Detail: The Distances Are Squared

UNIT 2

KEY CONCEPT · Why it matters, in one line each

It removes the minus signs a point 3 below the centre and one 3 above both count as being 3 away.

It punishes one bad point harder than several slightly-off ones. One point far away hurts a lot.

Which cuts both ways that is what makes k-Means precise - and what makes one strange value able to spoil a group.

The Centre of a Group Is Its Average

REAL-WORLD APPLICATION · **The balance point**

Take the group add up all the members' values, divide by how many there are. That is the centre.

It is called a centroid and it is a position, not one of your actual data points.

Nobody may be sitting on it just as the centre of a table is not a chair.

Why the Average, and Not Something Else?

KEY CONCEPT · This is where the name comes from

It is not a guess the average is provably the position with the smallest total squared distance to a set of points.

So the step is exact moving a centre to the average is the best possible move, every single time.

Hence the name k-MEANS. The name tells you what it does.

WORKSHEET Part A of the numerical worksheet proves this. Do it there.

How k-Means Runs - Two Steps, Over and Over

UNIT 2

Two steps you could do by hand, repeated until the answer stops changing.

01 START
- Drop k centres somewhere
- More on where in a moment

02 ASSIGN
- Every point joins its nearest centre
- That is the whole rule

03 UPDATE
- Each centre moves to the average
- of whatever joined it

04 REPEAT
- Assign again, update again
- Until nobody changes group

05 STOP
- Nothing moved, so nothing will
- The grouping has settled

CHECK Why must it stop eventually, rather than run for ever?

Where the Centres Start Changes the Answer

UNIT 2

COMMON PITFALL · The part beginners are most surprised by

Run it twice with different starting centres, and you can get two different groupings from the same data.

Neither is a bug both are genuine stopping points. One is simply better than the other.

So the answer depends partly on luck. That is a property of the method, not a mistake.

Two Ways to Choose the Starting Centres

Same algorithm afterwards. Only the starting positions differ.

	Random	k-Means++
How it picks	k data points at random	First at random, then favours far-away points
The risk	Two centres land in the same clump	Much less likely - they get spread out
What happens then	One real group is left with no centre	Each natural group tends to get one
Speed	Instant	Slightly slower to start, usually worth it
In practice	Rarely used alone	The default in most software

A Safety Net: Just Run It Several Times

EXAM TIP · What software does for you by default

The trick run the whole algorithm several times from different starts.

Then keep whichever run ended with the lowest inertia.

Why it works a bad start is unlucky. Being unlucky five times in a row is much rarer.

Limitation 1: You Have to Choose k

COMMON PITFALL · **And the algorithm will never tell you**

Ask for 4 groups you get 4. Ask for 9, you get 9.

Even from random noise it will hand back neat-looking groups that mean nothing at all.

Choosing k is your job and section 05 is entirely about how.

Limitation 2: Only Round Blobs

COMMON PITFALL · **It groups by 'nearest centre', so this follows**

The boundaries come out straight because every point simply joins whichever centre is closest.

So groups come out roughly round and roughly the same size.

A crescent or a ring cannot be found - the centre of a ring is not in the ring.

Limitation 3: One Odd Value Moves the Centre

COMMON PITFALL · **Because the centre is an average**

An average is easy to drag one very large value pulls it a long way.

Example average income in a room of ten people changes enormously if one billionaire walks in.

So the centre can end up describing nobody in the group.

Limitation 4: The Units You Recorded In Matter

COMMON PITFALL · **This one silently ruins real projects**

Distance adds up across columns so a column with big numbers contributes big differences.

Age in years, income in rupees income wins every time, and age might as well not be there.

Section 04 fixes it and it is the most common mistake in this unit.

But It Is Still the First Thing You Should Try

Every limitation above is real - and it is still the right first choice most of the time.

SPEED

It is fast

- Cost grows in a straight line with the data
- Handles very large datasets comfortably

SIMPLICITY

You can explain it

- Two steps, repeated
- You can describe it to a manager in a minute

USEFULNESS

It usually works

- On tidy, roughly round, scaled data it is hard to beat
- Start here, and move on only if it fails

03

SECTION

Two Variants That Fix Weaknesses

IN THIS SECTION

- k-Medoids, for data with odd values in it
- MiniBatch k-Means, for data that is very large

The Problem k-Medoids Solves

KEY CONCEPT · **Remember limitation 3 - the average gets dragged**

The average can sit nowhere in a group of 2, 3, 4, 5 and 100 the average is about 23.

Nobody is near 23 so the 'centre' describes none of the members.

Idea what if the centre had to be one of the actual members?

A Medoid Is a Real Member

REAL-WORLD APPLICATION · **Choosing a class representative**

You cannot invent one the representative has to be an actual student in the class.

You pick the most central one the student who is closest to everyone else on average.

That is a medoid a real data point chosen to stand for the group.

k-Means vs k-Medoids

One change - the centre must be a real data point - causes every difference below.

	k-Means	k-Medoids (PAM)
The centre is	An average - usually not a real data point	A real member of the group
Distances are	Squared	Not squared
One odd value	Drags the centre a long way	Barely moves it - the centre must stay on a real point
Speed	Fast	Much slower - it tests swapping members in and out
Use it when	Data is large and reasonably clean	Data has odd values, or the centre must be a real example

DECIDE Three million customers, and you know there are data-entry errors. Which do you pick?

The Problem MiniBatch Solves

KEY CONCEPT · **Nothing to do with messy data - this one is about size**

Every round, k-Means looks at everything all of it, to move the centres.

With a million rows that is a million checks per round, every round.

Idea what if each round only looked at a small random sample?

MiniBatch: Use a Sample Each Round

REAL-WORLD APPLICATION · **Like a survey instead of a census**

You do not ask everyone you ask a few thousand people and update your estimate.

The estimate wobbles a bit but it is close, and it is enormously faster.

Same trade here the centres take a slightly noisier path and end up slightly worse - for a fraction of the work.

04

SECTION

Scaling: The Units Problem

IN THIS SECTION

- Why the units you recorded in change the answer
- Two ways to put features on one scale
- When not to scale

Compare These Two People

KEY CONCEPT · **Age in years, income in rupees**

Person A 25 years old, earning 40,000.

Person B 45 years old, earning 41,000.

Ask the computer how different they are and it adds up the gaps: 20 years, and 1,000 rupees.

The Income Column Wins, Every Time

COMMON PITFALL · **And nobody told it to**

The age gap is 20 the income gap is 1,000. Fifty times larger, before anything is squared.

So the answer is decided by income and age has effectively no say at all.

Your 'age and income groups' turn out to be income bands with the same age range in every one.

Two Ways to Put Features on One Scale

Both put every column on comparable footing. They differ in how they handle odd values.

OPTION 1

Standardisation

- Re-write each value as how far it sits from that column's own average
- Measured in that column's own spread
- Odd values stay visible
- The usual default

OPTION 2

Min-max

- Stretch each column onto the range 0 to 1
- Smallest becomes 0, largest becomes 1
- Easy to explain
- But one odd value squashes everything else

When NOT to Scale

EXAM TIP · Scaling is not automatic - it is a decision

If the columns already match all of them counts of the same kind, in the same units - leave them alone.

If size IS the point and you want the big-spending customers to dominate, scaling would remove exactly what you care about.

Otherwise scale. And always scale before clustering, never after.

CHECK YOUR UNDERSTANDING

Quick Check

UNIT 2

QUICK CHECK

You cluster customers on age (18 to 70) and annual income (20,000 to 20,00,000) without scaling. What will your groups actually be?

- **A** A balanced mix of age and income groups
- **B** Income bands, with age contributing almost nothing
- **C** Age groups, because age has more distinct values
- **D** Unpredictable - it depends on k

ANSWER: B Income spans a range tens of thousands of times wider than age. Once distances are added up, income decides all of them. Age is in the model and has no influence. Your 'segments' are income bands.

05

SECTION

Checking the Answer

IN THIS SECTION

- Why you cannot just measure accuracy
- Inertia, and what it cannot do
- The elbow method, and where it lets you down
- The Davies-Bouldin Index

You Cannot Mark This Like an Exam

KEY CONCEPT · Back to the hard part from section 01

There is no answer key nobody knows the true grouping, so there is nothing to compare against.

So no accuracy, no marks out of ten those need a right answer to exist.

What you CAN measure the shape of the result. Are the groups tight? Are they well separated?

2.5 CLUSTER VALIDATION

Measure 1: Inertia

Useful, but it can never answer the question you most want to ask.

WHAT IT MEASURES

Tightness only

- How closely points sit around their own centre
- The same number k-Means was minimising

THE CATCH

It always falls

- Add a group and it can only get smaller
- Give every point its own group and it hits zero

SO

It cannot choose k for you

- Minimising it always says 'more groups'
- You can only look at it for a bend

The Elbow Method

REAL-WORLD APPLICATION · **Like deciding when to stop adding shelves**

The idea plot inertia as you increase the number of groups. It always falls.

Look for where it stops falling steeply the point where an extra group stops buying you much.

That bend is the 'elbow' and you read your k off it.

2.5 ELBOW METHOD PITFALLS

Four Ways the Elbow Lets You Down

A useful first look - not a decision procedure.

PROBLEM 1

Often there is no bend

- Real curves are frequently smooth
- Three people read three different answers

PROBLEM 2

It is a judgement call

- No rule defines where a bend is
- So the answer is not reproducible

PROBLEM 3

It ignores separation

- Inertia only knows about tightness
- It says nothing about groups being far apart

PROBLEM 4

It answers on noise

- Run it on structureless data
- You still get a curve, and still read an elbow

2.5 CLUSTER VALIDATION

Measure 2: Davies-Bouldin

UNIT 2

KEY CONCEPT · What inertia was missing

It looks at two things how spread out each group is, AND how far apart the groups sit.

Tight groups, far apart score low. Loose groups, close together, score high.

Low is good and zero would be perfect.

And It Can Actually Choose k

EXAM TIP · The thing inertia could not do

It does not just keep falling it gets worse if you add too many groups, as well as too few.

So it has a genuine best point you can try several values of k and pick the lowest score.

One catch each group is judged by its WORST neighbour - so one badly-placed group spoils the score.

Neither One Tells You It Is USEFUL

COMMON PITFALL · **The point to leave the unit with**

Both measure shape tight, separated, well-formed.

Neither measures meaning a beautifully tidy grouping can correspond to nothing anyone can act on.

That judgement is yours and no number will make it for you.

DISCUSS The elbow suggests 4 groups, Davies-Bouldin suggests 6, and marketing can run 3 campaigns. What do you deliver?

UNIT II CONSOLIDATION

Activity: Plan a Clustering

UNIT 2

GROUP WORK · 12 minutes

In groups of three or four, and on paper only - no code. You are making decisions and defending them.

1

Pick something to group

- Students by attendance and marks
- Shops by footfall and average basket
- Or anything with two or three numbers per row

2

Decide before you start

- Do the columns need scaling? Why?
- How many groups will you try?
- What would a useful group look like?

3

Then defend it

- Which measure would you check?
- What would make you say the result is not usable?

DELIVERABLE One person from each group: 60 seconds on what you chose to group, whether you scaled, and how you would know if the answer was any good.

UNIT II RECAP

Unit II - What to Take Away

UNIT 2

1. **Clustering finds groups nobody labelled**
   Numbers in, group labels out - and there is no answer key.

2. **k-Means makes groups tight**
   It measures total distance to each group's centre, and makes that number small.

3. **The centre is the average**
   Which is why it is called k-Means - and why one odd value can drag it.

4. **Where it starts changes what you get**
   So run it several times and keep the best. k-Means++ helps.

5. **Units decide the answer unless you scale**
   A big-numbered column takes over completely. Scale before clustering.

6. **k-Medoids for messy data, MiniBatch for big data**
   One makes the centre a real point; the other looks at a sample.

7. **Checking is about shape, not correctness**
   Inertia for tightness, Davies-Bouldin for tightness AND separation. Neither tells you it is useful.

NEXT UP Unit III - what to do when the groups are not round blobs: hierarchical clustering and DBSCAN.